

8. Read/ write the Local data.

1. Creating “8 program” Application.

1. Click Start → Android Studio, a Welcome to Android Studio dialog box will appear. Click New Project, the New Project Dialog box appears.

2. Choose Empty Views Activity then click Next.

3. Specify the Name of your project, Select the Language as Java, and Select the Minimum SDK as API 16 (“Jelly Bean”, Android 4.1). Click Finish Button.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
<EditText
    android:id="@+id/editTextData"
    android:layout_width="301dp"
    android:layout_height="48dp"
    android:layout_marginStart="99dp"
    android:layout_marginTop="81dp"
    android:layout_marginEnd="102dp"
    android:ems="10"
    android:hint="Enter some data"
    android:inputType="text"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
```

```
<Button
    android:id="@+id/buttonSave"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

```
android:layout_marginStart="161dp"
android:layout_marginTop="22dp"
android:layout_marginEnd="159dp"
android:text="Save"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/editTextData" />
```

<Button

```
android:id="@+id/butLoad"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="165dp"
android:layout_marginTop="22dp"
android:layout_marginEnd="158dp"
android:text="Load"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/buttonSave" />
```

<TextView

```
android:id="@+id/textViewData"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="176dp"
android:layout_marginTop="40dp"
android:layout_marginEnd="177dp"
android:layout_marginBottom="403dp"
android:text="Loaded data will appear here"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/butLoad" />
```

</androidx.constraintlayout.widget.ConstraintLayout>

Main_Activity.java

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.content.SharedPreferences;
```

```
import android.os.Bundle;
```

```
import android.view.View;
```

```
import android.widget.Button;
```

```
import android.widget.EditText;
```

```
import android.widget.TextView;
```

```
import android.widget.Toast;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    private EditText editTextData;
```

```
    private Button buttonSave, buttonLoad;
```

```
    private TextView textViewData;
```

```
    private SharedPreferences sharedPreferences;
```

```
    private static final String SHARED_PREF_NAME = "MyAppPrefs";
```

```
    private static final String KEY_DATA = "savedData";
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        editTextData = findViewById(R.id.editTextData);
```

```
        buttonSave = findViewById(R.id.buttonSave);
```

```
        buttonLoad = findViewById(R.id.butLoad);
```

```
        textViewData = findViewById(R.id.textViewData);
```

```
        sharedPreferences = getSharedPreferences(SHARED_PREF_NAME, MODE_PRIVATE);
```

```
        buttonSave.setOnClickListener(new View.OnClickListener() {
```

```
            @Override
```

```
        public void onClick(View v) {

            saveData();

        }

    });

    buttonLoad.setOnClickListener(new View.OnClickListener() {

        @Override

        public void onClick(View v) {

            loadData();

        }

    });

}

private void saveData() {

    String data = editTextData.getText().toString().trim();

    if (!data.isEmpty()) {

        SharedPreferences.Editor editor = sharedPreferences.edit();

        editor.putString(KEY_DATA, data);

        editor.apply();

        Toast.makeText(this, "Data saved successfully", Toast.LENGTH_SHORT).show();

    } else {

        Toast.makeText(this, "Please enter some data", Toast.LENGTH_SHORT).show();

    }

}

private void loadData() {

    String data = sharedPreferences.getString(KEY_DATA, "");

    textViewData.setText(data);

    Toast.makeText(this, "Data loaded successfully", Toast.LENGTH_SHORT).show();

}

}
```

